Machine Learning 2024-2045

Homework - Group 127

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I. **- Pen-and-Paper**

(1) • $IG(Y_{out} \mid Y_1 \geq 0.3, Y_i) = E(Y_{out} \mid Y_1 \geq 0.3) - E(Y_{out} \mid Y_1 \geq 0.3, Y_i)$

• $E(Y_{out} \mid Y_1 \geq 0.3) = -\frac{3}{7}\log_2\left(\frac{3}{7}\right) - \frac{2}{7}\log_2\left(\frac{2}{7}\right) - \frac{2}{7}\log_2\left(\frac{2}{7}\right) \simeq 1,5566.$

• $E(Y_{out} \mid Y_1 \geq 0.3, Y_2) = 4/7\, E(Y_{out} \mid Y_1 \geq 0.3, Y_2 = 0) + \frac{3}{7} E(Y_{out} \mid Y_1 \geq 0.3, Y_2 = 1)$
$+ \frac{0}{7} E(Y_{out} \mid Y_1 \geq 0.3, Y_2 = 2) = 4/7\left(-\frac{1}{4}\log_2\left(\frac{1}{4}\right) - \frac{1}{4}\log_2\left(\frac{1}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{4}\right)\right) +$
$+ \frac{3}{7}\left(-\frac{2}{3}\log_2\left(\frac{2}{3}\right) - \frac{1}{3}\log_2\left(\frac{1}{3}\right) - 0\cdot\log_2(0)\right) = 4/7 \cdot \frac{3}{2} + \frac{3}{7} \cdot \frac{\log_2\left(\frac{27}{4}\right)}{3} \simeq$
$\simeq 1,2507.$

• $E(Y_{out} \mid Y_1 \geq 0.3, Y_3) = \frac{2}{7} E(Y_{out} \mid Y_1 \geq 0.3, Y_3 = 0) + 4/7\, E(Y_{out} \mid Y_1 \geq 0.3, Y_3 = 1)$
$+ 1/7\, E(Y_{out} \mid Y_1 \geq 0.3, Y_3 = 2) = \frac{2}{7}\left(-1\log_2(1) - 0\log_2(0) - 0\log_2(0)\right) +$
$+ 4/7\left(-1/4\log_2\left(\frac{1}{4}\right) - \frac{1}{4}\log_2\left(\frac{1}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{4}\right)\right) + 1/7\left(-0\log_2(0) - 1\log_2(1) - 0\log_2(0)\right)$
$= \frac{2}{7}\times 0 + 4/7 \times \frac{3}{2} + \frac{1}{7}\times 0 = 6/7 \simeq 0,857.$

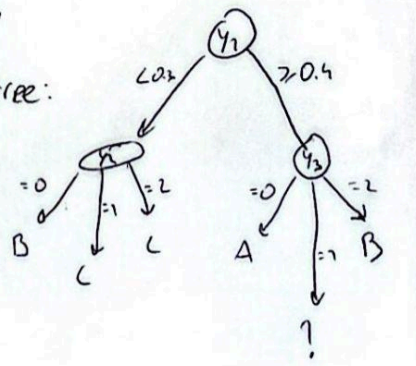
• $E(Y_{out} \mid Y_1 \geq 0.3, Y_4) = 4/7\, E(Y_{out} \mid Y_1 \geq 0.3, Y_4 = 0) + \frac{3}{7} E(Y_{out} \mid Y_1 \geq 0.3, Y_4 = 1)$
$+ \frac{0}{7} E(Y_{out} \mid Y_1 \geq 0.3, Y_4 = 2) = 4/7\left(-\frac{2}{4}\log_2\left(\frac{2}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{4}\right) - 0\log_2(0)\right) +$
$+ \frac{3}{7}\left(-\frac{1}{3}\log_2\left(\frac{1}{3}\right) - 0\log_2(0) - \frac{2}{3}\log_2\left(\frac{2}{3}\right)\right) = 4/7 \times 1 + \frac{3}{7} \cdot \frac{\log_2\left(\frac{27}{4}\right)}{3} \simeq 0,9650.$

• $IG(Y_{out} \mid Y_1 \geq 0.3, Y_2) = 1,557 - 1,2507 = 0,3060.$

• $IG(Y_{out} \mid Y_1 \geq 0.3, Y_3) = 1,557 - 0,857 = 0,6995.$

• $IG(Y_{out} \mid Y_1 \geq 0.3, Y_4) = 1,557 - 0,9650 = 0,5917.$

• Como Y_3 é o maior, vamos optar por usar Y_3 na tree:



• ? → Como temos o número mínimo de observações necessárias (4), podemos continuar a expandir este nó:

- $IG(Y_{out} \mid Y_1 \geq 0.3, Y_3 = 1, Y_i) = E(Y_{out} \mid Y_1 \geq 0.3, Y_3 = 1) - E(Y_{out} \mid Y_1 \geq 0.3, Y_3 = 1, Y_i)$
- $E(Y_{out} \mid Y_1 \geq 0.3, Y_3 = 1) = \left(-1/4 \log_2(\frac{1}{4}) - 1/4 \log_2(\frac{1}{4}) - \frac{2}{4} \log_2(\frac{2}{4})\right) = 1{,}5.$
- $E(Y_{out} \mid Y_1 \geq 0.3, Y_3 = 1, Y_2) = 4/4\, E(Y_{out} \mid Y_1 \geq 0.3, Y_3 = 1, Y_2 = 0) +$

  $+\, 0 \cdot E(Y_{out} \mid Y_1 \geq 0.3, Y_3 = 1, Y_2 = 1) + 0 \cdot E(Y_{out} \mid Y_1 \geq 0.3, Y_3 = 1, Y_2 = 2)$

  $= 1 \cdot \left(-\frac{1}{4} \log_2(\frac{1}{4}) - \frac{1}{4} \log_2(\frac{1}{4}) - \frac{2}{4} \log_2(\frac{2}{4})\right) = 1{,}5.$
- $E(Y_{out} \mid Y_1 \geq 0.3, Y_3 = 1, Y_4) = 1/4\, E(Y_{out} \mid Y_1 \geq 0.3, Y_3 = 1, Y_4 = 0) +$

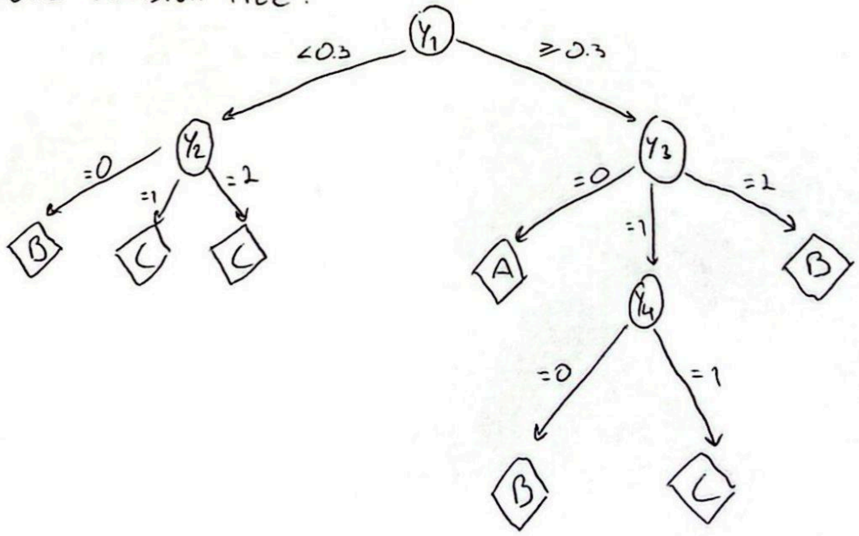
  $+\, \frac{3}{4} E(Y_{out} \mid Y_1 \geq 0.3, Y_3 = 1, Y_4 = 1) + 0 \cdot E(Y_{out} \mid Y_1 \geq 0.3, Y_3 = 1, Y_4 = 2) =$

  $= \frac{1}{4}\left(-0 \cdot \log_2(0) - \frac{1}{1} \log_2(1) - 0 \cdot \log_2(0)\right) + \frac{3}{4}\left(-\frac{1}{3} \log_2(\frac{1}{3}) - 0 \cdot \log_2(0) - \frac{2}{3} \log_2(\frac{2}{3})\right)$

  $= \frac{1}{4} \cdot 0 + \frac{3}{4} \cdot \frac{\log_2(\frac{27}{4})}{3} = 0{,}6887.$
- $IG(Y_{out} \mid Y_1 \geq 0.3, Y_3 = 1, Y_2) = 1{,}5 - 1{,}5 = 0.$
- $IG(Y_{out} \mid Y_1 \geq 0.3, Y_3 = 1, Y_4) = 1{,}5 - 0{,}6887 = 0{,}8113.$

Logo, optamos por Y_4.

Complete Decision Tree:



TÉCNICO LISBOA

2)

	Prediction $\hat{z}$	Real z
x_1	C	C
x_2	B	B
x_3	C	C
x_4	B	B
x_5	C	C
x_6	B	B
x_7	C	A
x_8	A	A
x_9	C	C
x_{10}	C	C
x_{11}	A	A
x_{12}	B	B

Training Confusion Matrix:

Prediction \ Real	A	B	C	Total
A	2	0	0	2
B	0	4	0	4
C	1	0	5	6
Total	3	4	5	

3) $F\text{-measure} = \dfrac{1}{\alpha \cdot \frac{1}{P} + (1-\alpha)\frac{1}{R}}$ $\quad F1 = \frac{1}{2}\left(\frac{1}{P} + \frac{1}{R}\right) = \dfrac{2 \times P \times R}{P+R}$

$\beta^2 = \dfrac{1-\alpha}{\alpha}$ (F1-measure com $\beta = 1$), logo $\alpha = 1/2$

P → PRECISION
R → RECALL

$\cdot\, Precision_A = \dfrac{TP_A}{TP_A + FP_A} = \dfrac{2}{2+0} = 1$ $\quad \cdot\, Recall_A = \dfrac{TP_A}{FN_A + TP_A} = \dfrac{2}{2+1} = \dfrac{2}{3}$ $\quad \cdot\, F1_A = \dfrac{2 \cdot \frac{2}{3} \times 1}{1 + \frac{2}{3}} = 4/5$

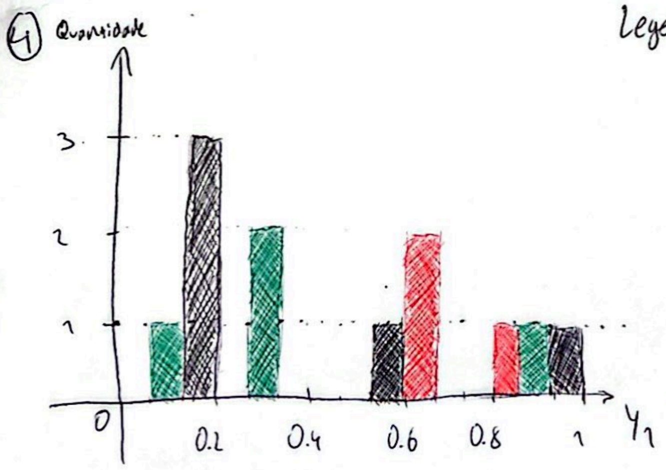
$\cdot\, Precision_B = \dfrac{TP_B}{TP_B + FP_B} = \dfrac{4}{4+0} = 1$ $\quad \cdot\, Recall_B = \dfrac{TP_B}{FN_B + TP_B} = \dfrac{4}{4+0} = 1$ $\quad \cdot\, F1_B = \dfrac{2 \cdot 1 \cdot 1}{1+1} = 1$

$\cdot\, Precision_C = \dfrac{TP_C}{TP_C + FP_C} = \dfrac{5}{5+0} = 1$ $\quad \cdot\, Recall_C = \dfrac{TP_C}{TP_C + FN_C} = \dfrac{5}{5+1} = \dfrac{5}{6}$ $\quad \cdot\, F1_C = \dfrac{2 \times 1 \times \frac{5}{6}}{1 + 5/6} = \dfrac{10}{11}$

Temos $F1_B > F1_C \simeq 0{,}9091 > F1_A \simeq 0{,}8$, logo, a classe com menor valor de F1 é a classe A.

4)

Legenda: A, vermelho
B, verde
C, preto



Divisão da raiz:

$\cdot\,]0\,;0.2[\rightarrow C$

$\cdot\,]0.2\,;0.6[\rightarrow B$

$\cdot\,]0.6\,;1[\rightarrow A$